7.3 and 7.4: Basis Functions and Regression Splines

Basis Functions

- Polynomial and piece-wise constant regression models are a special case of a *basis function* approach.
- Idea: use a family of functions or transformations that can be applied to a variable X:

$$b_1(X), b_1(X), \ldots, b_K(X)$$

  and then we fit the model

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \beta_3 b_3(x_i) + \cdots + \beta_K b_K(x_i) + \epsilon_i$$

- These basis functions b are fixed and known (chosen in advance).
- Like polynomial regression, these basis functions just create a new input matrix which can be used in a standard linear model.
  - So all of our linear regression inference tools apply!

Regression Splines: Piecewise Polynomials

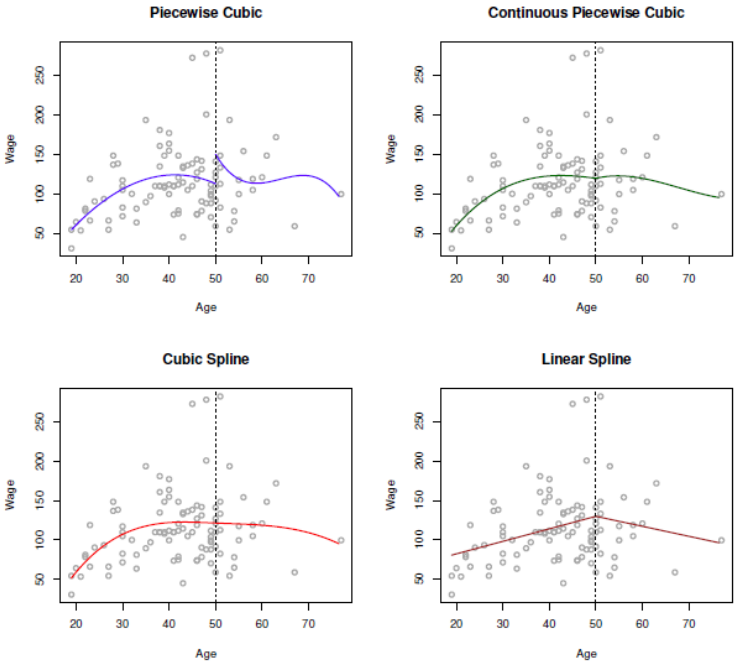
- Instead of a single polynomial in X over its whole domain, w can use different polynomials defined by knots.
- Ex:

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if} \quad x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if} \quad x_i \geq c \end{cases}$$

- Downside: piecewise approaches can create bizarre discontinuities.

Constraints and Splines

- In general, it's better to add some constraints to the polynomials.
  - One such constraint is continuity at the knots.
- *Splines* have the maximum amount of continuity.
- Each constraint *increases* degrees of freedom $+1$ by reducing the complexity of the fit.


Piecewise Cubic
Continuous Piecewise Cubic
Cubic Spline
Linear Spline
Wage
Age
50
100
150
200
250
20
30
40
50
60
70

Constraints and Splines

- Top left: no constraint on the cubic polynomials.
- Top right: polynomials forced to be continuous at the knot.
  - $+1$ degree of freedom
- Bottom left: both first and second derivatives are continuous.
  - $+2$ degrees of freedom
- Bottom right: linear spline continuous at knot.

The Spline Basis Representation

- We can use the basis approach to represent regression splines.
- A linear spline with K knots can be modeled as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \epsilon_i$$

The Spline Basis Representation

We start with the basis for a linear model, then add one *truncated power basis* function per knot

$$
\begin{aligned}
b_1(x_i) &= x_i \\
b_{k+1}(x_i) &= (x_i - \xi_k)_+, \quad k = 1, \ldots, L
\end{aligned}
$$

where ξ is the knot and $(x)_+$ means the *positive part* of x:

$$
(x)_+ = \begin{cases} x_i & \text{if} \quad x > 0 \\ 0 & \text{otherwise} \end{cases}
$$

Cubic Splines

A cubic spline with knots at ξ_k, $k = 1, \ldots, K$ is a piecewise cubic polynomial with continuous derivatives up to order 2 at each knot.

We can represent this as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$$

where now the truncated basis function used is

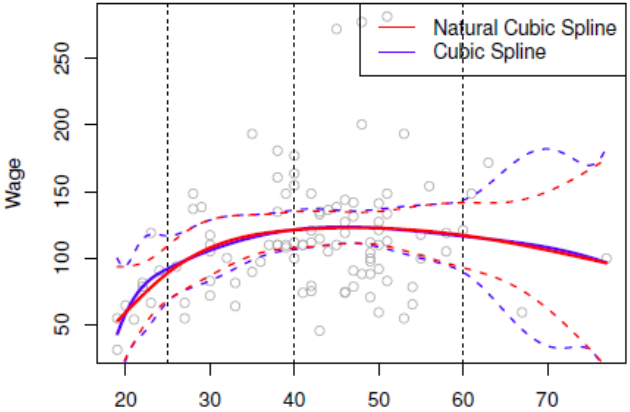
$$h(x, \xi) = (x - \xi)_+^3$$

Cubic Splines

- This amounts to fitting a model with predictors of the form $X, X^2, X^3, h(X, \xi_1), h(X, \xi_2), \ldots, h(X, \xi_K)$
  - $\xi_1, \ldots, \xi_K$ are the knots.
  - There are $K + 4$ regression coefficients (including the intercept), so the model uses $K + 4$ degrees of freedom.

Natural Cubic Splines

- Splines can have high variance at the outer range of X
- Natural splines have additional boundary constraints (linearity at boundaries)
- This adds $4 = 2 \times 2$ additional constraints.
- Estimates generally more stable at the boundaries

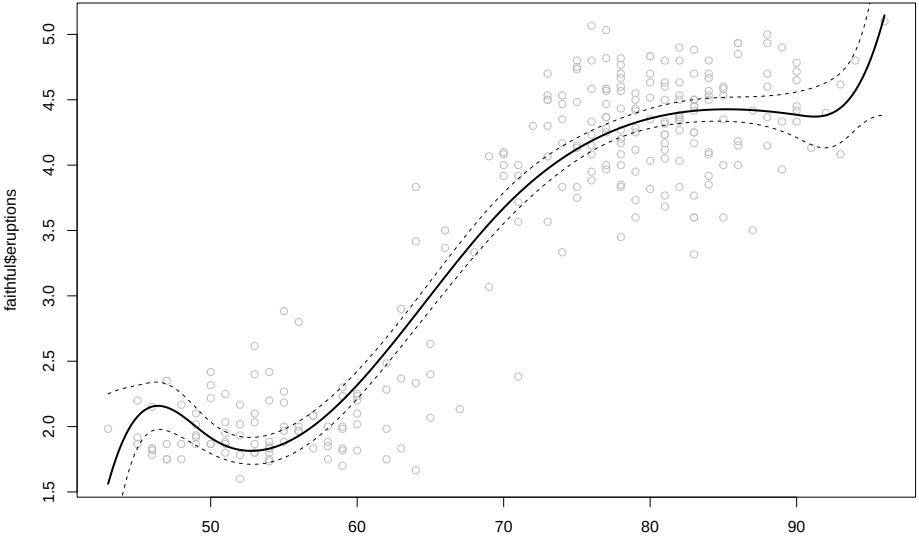


Choosing the Knots

The number and location of knots needs to be set. How?

- Model most flexible in regions with many knots.
  - So we might place more knots in places where the function might vary most rapidly.
- In practice, knots are typically placed uniformly.
  - Specify desired degrees of freedom and place knots at uniform quantiles.

Regression Splines in R

```r
library(splines)
fit <- lm(eruptions ~ bs(waiting, knots = c(50, 70, 90)), data = faithful)
wait.grid <- seq(43,96,0.1)
pred <- predict(fit, newdata = list(waiting = wait.grid), se = T)
plot(faithful$waiting, faithful$eruptions, col = "gray")
lines(wait.grid, pred$fit, lwd = 2)
lines(wait.grid, pred$fit + 2 * pred$se, lty = "dashed")
lines(wait.grid, pred$fit - 2 * pred$se, lty = "dashed")
```

Regression Splines in R